

HR Workforce Analytics Dashboard

Business Intelligence & Workforce Insights Report

Power BI | Data Modeling | DAX | Power Query

Transforming Workforce Data into Actionable Insights

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HR Workforce Analytics Dashboard

An end-to-end HR analytics project built in Power BI to help management understand workforce distribution, salary patterns, leave behaviour, overtime load, and employee growth across countries and departments.

Executive Summary

This project turns a flat employee dataset into an executive dashboard with clear HR KPIs and decision-ready insights.

The analysis shows:

- A workforce of **689 employees** across **20 departments** and **5 countries**.
- **Manufacturing** is the largest department, while **Egypt** has the largest employee base.
- Employee growth peaked in **2019**, with the largest year-over-year workforce increase.
- Average salary is **24.8K** annually, with visible variation across departments and countries.
- Average sick leave is **1.61 days** per employee, and the relationship between overtime and sick leave is essentially negligible in this dataset.

Business Problem

The HR team lacked a centralized view of workforce metrics, which made it difficult to monitor employee trends, compare departments, and support strategic decisions around staffing, compensation, and workload planning.

Project Objectives

- Monitor workforce KPIs in one place.
- Analyse employee distribution by department, country, and gender.
- Track salary and leave patterns.
- Identify workforce trends over time.
- Highlight areas that require management attention.

Dataset Snapshot

- Rows: **689 employees**
- Columns: **15**
- Missing values: **0**
- Duplicate rows: **0**
- Time span: **2016 to 2020**
- Countries: **Egypt, United Arab Emirates, Saudi Arabia, Syria, Lebanon**
- Departments: **20**

KPI Summary

- Total employees: **689**
- Average performance rating: **3.6**
- Average sick leaves per employee: **1.61**
- Average unpaid leaves per employee: **0.76**
- Average overtime hours per employee: **13.70**
- Total annual salary: **17.10M**
- Average annual salary: **24.8K**

Key Insights

- 1. Manufacturing leads headcount**
 - a. Manufacturing has the largest workforce with **140 employees**.
 - b. This makes it the most important area for staffing, scheduling, and operational planning.
- 2. Egypt dominates workforce concentration**
 - a. Egypt has **379 employees**, the highest count among all countries.
 - b. That concentration should be considered when interpreting country-level salary and overtime patterns.
- 3. Workforce growth peaked in 2019**
 - a. Headcount rose from **51 in 2016** to **248 in 2019**, then declined to **162 in 2020**.
 - b. This pattern suggests either a hiring slowdown, restructuring, or a change in staffing strategy after the 2019 peak.
- 4. Compensation varies materially by department**
 - a. Highest average annual salary: **Human Resources** at about **30.7K**
 - b. Lowest average annual salary: **Research Center** at about **22.6K**

- c. This gap is meaningful and worth reviewing for internal equity and role alignment.
- 5. **Overtime and sick leave are not meaningfully linked here**
 - a. The correlation between overtime hours and sick leaves is approximately **-0.03**.
 - b. In this dataset, overtime does not appear to explain sick leave behaviour.
- 6. **2020 employees show the highest average salary**
 - a. Employees who started in **2020** have the highest average annual salary in the dataset.
 - b. That may indicate a changed hiring mix, revised pay structure, or both.

Recommended Actions

- Review salary bands by department to check for consistency and internal equity.
- Investigate the 2019 to 2020 workforce drop to understand whether it was driven by attrition, restructuring, or hiring policy changes.
- Segment leave analysis by department and country to identify localized absenteeism patterns.
- Add turnover, tenure, and promotion data in a future version to make the dashboard more strategic.
- Convert the current descriptive dashboard into a more diagnostic HR performance model with drill-through pages and trend comparisons.

Dashboard Pages

- **Business Problem** - explains the context and challenges.
- **Business Requirements** - documents the questions the dashboard is designed to answer.
- **HR Employees Report** - headcount, gender, country, leave, overtime, and growth analysis.
- **Salaries Employees Report** - salary distribution by year of experience, country, department, and overtime.
- **Diagnostic Details** - highlights workload pressure, leave patterns, performance alignment, and payroll concentration by department.
- **Summary Report** - executive-level narrative summary of the main findings.

Tools Used

- Data Cleaning & Transformation
- Data Modelling
- Interactive Dashboard Design
- KPI Development
- Business Insights Generation

Limitations

- The dataset is a snapshot, so it does not include turnover, exit reasons, or promotion history.
- Correlations shown in the dashboard should not be treated as causal relationships.
- Country comparisons should be interpreted carefully because the sample is uneven across locations.

Dashboard Preview

Business Problem

The business challenge and the reason for building the dashboard.

Business Problem

Background

The HR department manages employees across multiple departments and countries. However, management lacks a centralized view of workforce distribution, employee performance, leave patterns, and salary allocation. As the company continues to grow, decision-makers need data-driven insights to better understand workforce trends and support strategic HR planning.

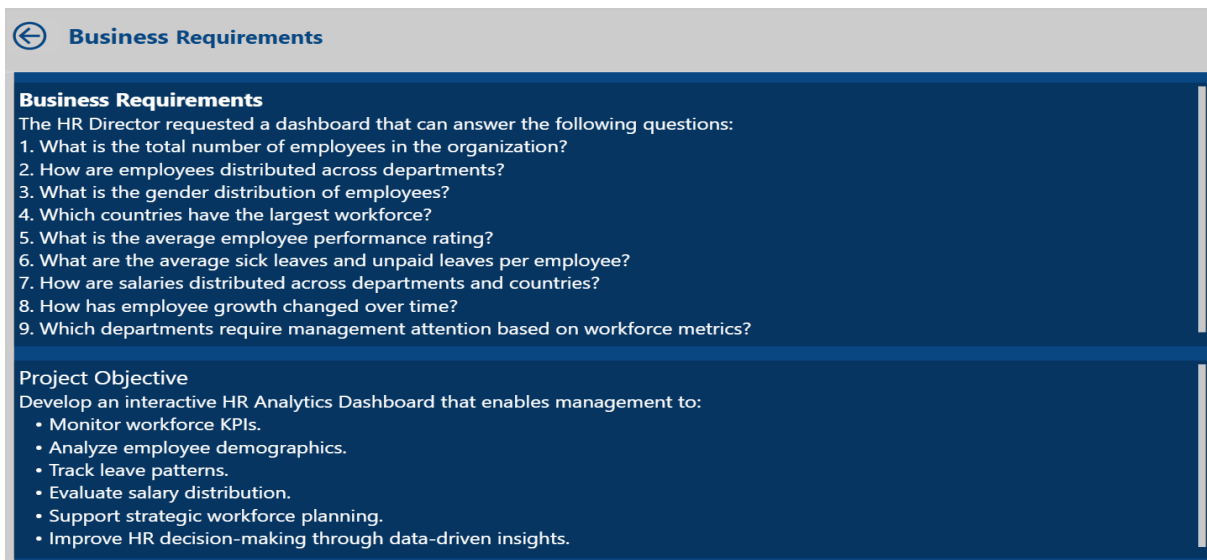
Business Challenges

The HR management team faces several challenges:

- Limited visibility into employee distribution across departments and countries.
- Difficulty monitoring employee performance and job ratings.
- Lack of insight into sick leaves and unpaid leaves.
- Challenges in evaluating salary distribution and compensation fairness.
- Inability to quickly identify workforce trends over time.

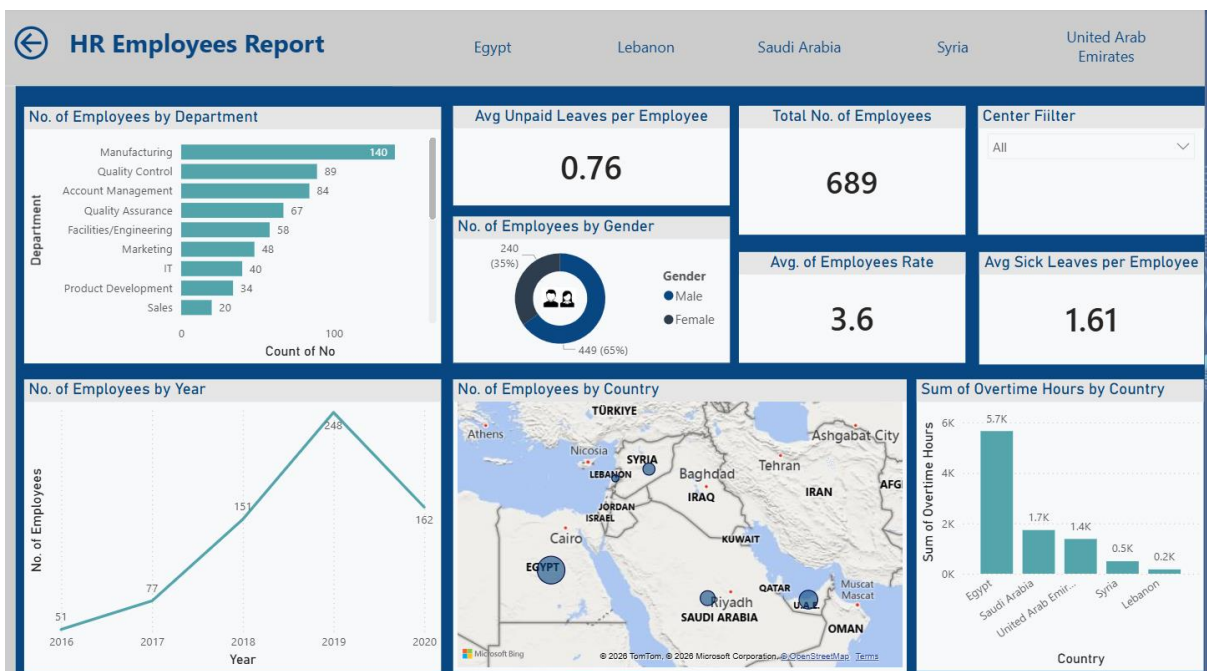
Business Requirements

The key questions the dashboard is designed to answer.



HR Employees Report

Headcount, country, gender, leave, overtime, and workforce growth analysis.



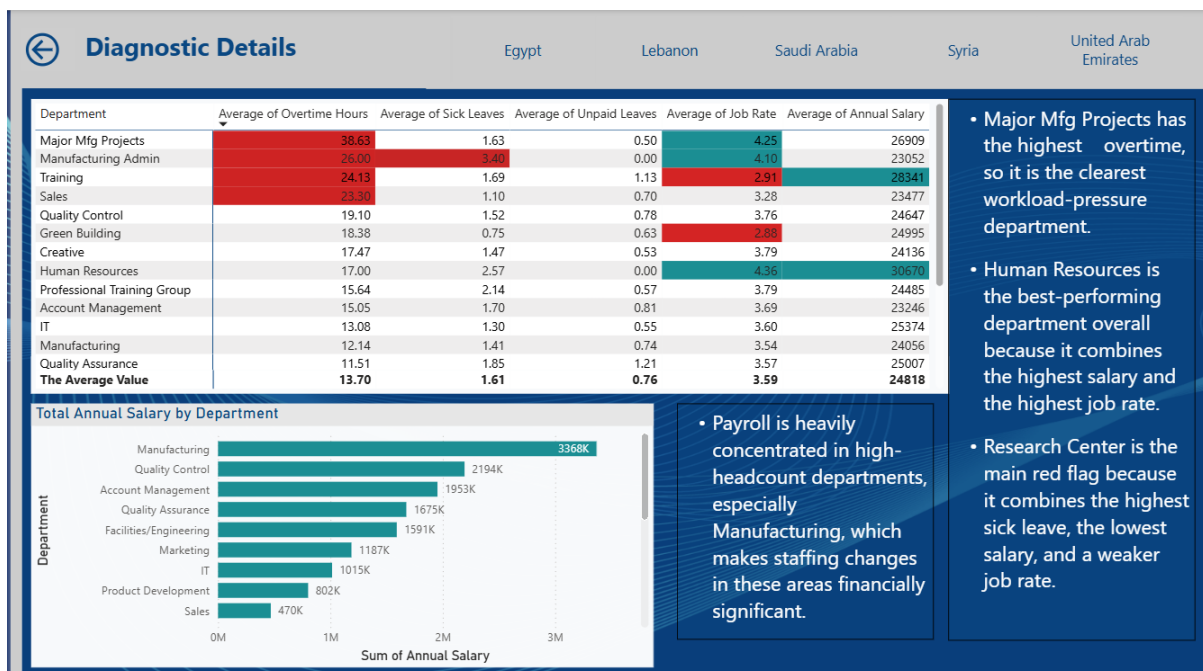
Salaries Employees Report

Salary distribution across departments, countries, and years of experience.



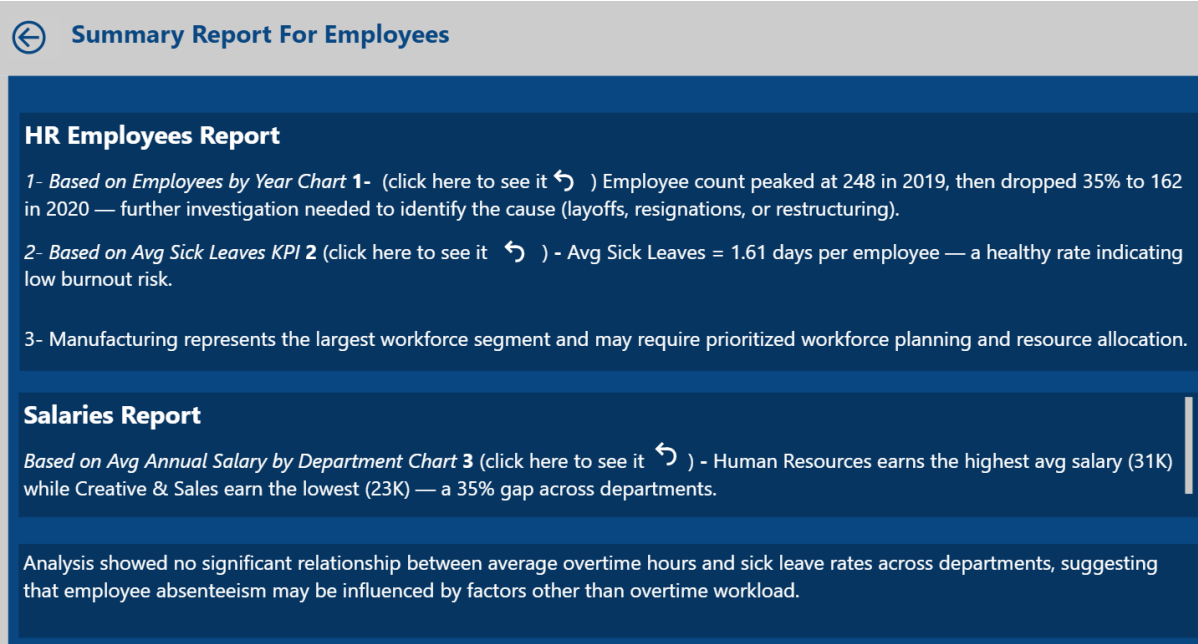
Diagnostic Details

Department-level workload, leave, performance, and payroll comparison.



Summary Report

Executive-level narrative of the main findings.



Why This Project Matters

This dashboard demonstrates the ability to transform raw HR data into actionable business insights. It combines data cleaning, modelling, KPI development, and interactive visualization to support data-driven decision-making.

GitHub Link: <https://github.com/ebrahimqlini/HR-Analytics-Dashboard/tree/main>